

USE-CASE FLOW SPECIFICATION

Use Case: Confirm Carpool

Primary Actor: Rider / Commuter

Goal: Confirm a suitable carpool trip and calculate the rider's fuel cost share.

Preconditions

1. Rider is registered and logged in.
2. A driver has created an available trip.
3. A compatible route is available within 1.5 km detour tolerance.

Main Success Scenario

1. Rider enters route and travel time.
2. System searches for available trips.
3. System checks route compatibility.
4. System displays compatible trips.
5. Rider selects a trip.
6. System displays trip details.
7. Rider confirms the carpool.
8. System records the rider.
9. System calculates the fuel cost share.
10. System displays the cost share.
11. System confirms successful participation.

Alternate Flow – Route Exceeds Tolerance

1. System detects that the required detour exceeds 1.5 km.
2. System rejects the route as incompatible.
3. System displays “No compatible carpool found.”
4. Rider may search again with different details.

Postconditions

1. Rider is successfully added to the carpool.
2. Fuel cost share is calculated and displayed.

